

## HW 1

1) A service center for kitchen appliances has hired a large group of new technicians. During the first 20 weeks of their employment, data were collected documenting the number of appliances serviced by a random sample of these technicians.

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Week   | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Number | 12 | 21 | 18 | 13 | 28 | 20 | 29 | 35 | 26 | 20 | 36 | 40 | 28 | 52 | 32 | 32 | 55 | 56 | 49 | 60 |

Here is part of the Minitab regression output for these data:

|                  |               |               |             |              |
|------------------|---------------|---------------|-------------|--------------|
| <b>Predictor</b> | <b>Coef</b>   | <b>StDev</b>  | <b>T</b>    | <b>P</b>     |
| <b>Constant</b>  | <b>10.679</b> | <b>3.554</b>  | <b>3.00</b> | <b>0.008</b> |
| <b>Week</b>      | <b>2.1353</b> | <b>0.2967</b> | <b>7.20</b> | <b>0.000</b> |

**S = 7.651      R-Sq = 74.2%      R-Sq(adj) = 72.8%**

(a) Interpret the slope of the regression in the context of the problem.

(b) What is the equation for the LSRL line?

(c) Using the Minitab output, determine a 99% confidence interval for the true slope of the regression line. Just carry out the calculation

(d) Interpret the meaning of the confidence interval in the context of the problem. What does this interval suggest about the employment of the new technicians?

## Multiple Choice Practice

\_\_\_\_\_ 2) Inference for regression on the population regression slope  $\beta$  is based on which of the following distributions?

- (A) The  $t$  distribution with  $n - 1$  degrees of freedom
- (B) The standard Normal distribution
- (C) The chi-square distribution with  $n - 1$  degrees of freedom
- (D) The  $t$  distribution with  $n - 2$  degrees of freedom
- (E) The Normal distribution with mean  $\mu$  and standard deviation  $\sigma$

Questions 3 to 4 refer to the following situation:

One concern about the depletion of the ozone layer is that the increase in ultraviolet light will decrease crop yields. An experiment was conducted in a greenhouse where soybean plants were exposed to varying levels of UV rays—measured in Dobson units. At the end of the experiment the yield (kg) was measured. A regression analysis was performed; here is some output:

| Parameter Estimates |           |           |            |         |                  |           |
|---------------------|-----------|-----------|------------|---------|------------------|-----------|
| Term                | Estimate  | Std Error | t Ratio    | Prob> t | Lower 95%        | Upper 95% |
| Intercept           | 3.9800118 | 0.053774  | 74.01      | <.0001  | 3.8638398        | 4.0961838 |
| uv                  | -0.046285 | 0.010741  | * hidden * | 0.0008  | **** hidden **** |           |

\_\_\_\_\_ 3) Which of the following is correct?

- (A) If the UV reading is increased by 1 Dobson unit, the yield is expected to increase by 0.0463 kg.
- (B) If the yield increases by 1 kg, the UV reading is expected to decline by 0.0463 Dobson units.
- (C) The estimated yield is 3.98 kg when the UV reading is 0 Dobson units.
- (D) The predicted yield is 4.3 kg when the UV reading is 20 Dobson units.
- (E) The  $t$  ratios are used to test if the estimated slope is different from zero.

\_\_\_\_\_ 4) A 95% confidence interval for the slope will be centered on the estimated slope and

- (A)  $\pm 0.011$ .      (B)  $\pm 0.108$ .      (C)  $\pm 0.054$ .      (D)  $\pm 0.046$ .      (E)  $\pm 0.021$ .

Questions 5 to 6 refer to the following situation:

Does drinking large amounts of high-sugar soft drinks play a significant role in being overweight? A random sample of 30 adult men were asked how much non-diet soda they drank on a typical day. Their body-mass index—a measure that is higher for overweight people—was also calculated. The computer output on the next page summarizes a regression analysis of the response variable body-mass index (BMI) and the explanatory variable “soda” (ounces of non-diet soft drinks per day).

| Predictor | Coef    | SE Coef | T     | P     |
|-----------|---------|---------|-------|-------|
| Constant  | 25.043  | 1.858   | 13.48 | 0.000 |
| Soda      | 0.07291 | 0.03863 | 1.89  | 0.070 |

$S = 6.96070$      $R\text{-Sq} = 11.3\%$      $R\text{-Sq(adj)} = 8.1\%$

\_\_\_\_\_ 5) Assuming the conditions for inference have been met, which of the following represents the 95% confidence interval for the slope of the population regression line relating BMI to soda consumption?

- (A)  $0.07291 \pm 2.048(0.03863)$
- (B)  $0.07291 \pm 2.048\left(\frac{0.03863}{\sqrt{28}}\right)$
- (C)  $0.07291 \pm 2.048\left(\frac{0.03863}{\sqrt{30}}\right)$
- (D)  $0.07291 \pm 1.96(0.03863)$
- (E)  $0.07291 \pm 1.96\left(\frac{0.03863}{\sqrt{30}}\right)$

\_\_\_\_\_ 6) Which of the following is a correct interpretation of “95% confidence” in reference to the interval (a, b)?

- (A) In 95% of repeated samples, the slope of sample regression lines for the regression of BMI on soda will fall between a and b.
- (B) In 95% of repeated samples, the true slope of the population regression line for the relationship between BMI and soda will fall between a and b.
- (C) The probability that the true slope of the population regression line for the relationship between BMI and soda is in the interval is 0.95.
- (D) The method used to construct this confidence interval will produce an interval from a to b 95% of the time.
- (E) The method used to construct this confidence interval will produce an interval that captures the true slope of the population regression equation 95% of the time.